MBS-003 - Why Dreaming Animals Matter
Dreaming as Evolutionary Evidence for MET Brain Structure

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with chatGPT (OpenAI)
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Repository: https://github.com/sizhet/MET-Brain-Structure-MBS

Introduction
One of the most fascinating questions in the study of intelligence is not why humans dream.
A more fundamental question may be:
Why do animals dream?
Humans dream.
Dogs appear to dream.
Cats appear to dream.
Many mammals exhibit dream-like sleep behavior.
Birds also show evidence of sleep states associated with memory replay and neural reactivation.
These observations are remarkable because dreaming appears to exist far below the level of language, mathematics, symbolic reasoning, scientific thought, or civilization itself.
Dreaming is ancient.
Evolutionarily speaking, dreaming existed long before philosophy, long before writing, and perhaps long before what we normally call human intelligence.
For the MET Brain Structure (MBS) project, this observation carries profound implications.
If dreaming is widespread across species, then dreaming may reveal something fundamental about the architecture of biological intelligence.
Rather than being a mysterious side effect, dreaming may expose one of the core runtime mechanisms that biological cognition has relied upon for millions of years.

The Importance of Removing Human Exceptionalism
Many theories of intelligence begin from uniquely human abilities:
- Language

- Logic
- Mathematics
- Planning
- Culture
- Technology

While these abilities are important, they can also obscure deeper biological principles.
Dreaming animals provide a useful counterexample.
A dog does not possess formal mathematics.
A dog does not study logic.
A dog does not write language.
Yet dogs appear capable of dream-like experiences.
During sleep, dogs may exhibit:

- Running motions
- Paw movements
- Vocalizations
- Sniffing behaviors
- Tracking-like reactions

These behaviors strongly suggest activation of internal scene-related processes.
If such processes exist without language or symbolic reasoning, then their foundations must be more primitive and more fundamental than human intellectual systems.
This immediately shifts attention toward embodied cognition.

Dreaming Suggests an Internal Scene Runtime

Suppose a dog dreams about chasing prey.
The dream may not be a perfect replay of reality.
However, it appears to involve several essential components:

- Objects
- Environments
- Actions
- Goals
- Transitions
- Behavioral sequences

In structural terms, this implies the existence of:

- State representations
- Action pathways
- Trigger mechanisms
- Runtime activation systems

In other words, dreaming appears to require some form of internal scene construction and execution.
The exact biological implementation remains unknown.
However, from a structural perspective, dreaming suggests the existence of a runtime architecture capable of activating meaningful embodied experiences without direct external input.
This observation aligns naturally with the MET Brain Structure hypothesis.

Dreaming and Calling Graphs

Within Structural Intelligence research, Calling Graphs are used to represent transitions among states through operations.
A simplified form can be expressed as:
State → Operation → State
Dreaming appears surprisingly compatible with this structure.
For example:
Resting State
→ Detection of Relevant Memory
→ Hunting Scene Activation
Hunting Scene
→ Pursuit Behavior
→ Capture Attempt
Capture Attempt
→ Reward Expectation
→ Scene Transition
The biological implementation may be entirely different from software systems.
However, the structural pattern appears similar.
Dreaming therefore provides a possible bridge between biological cognition and graph-based runtime models.

Dreaming and Function Tunnel Networks

The same observation can be viewed through Function Tunnel Networks.
A hunting sequence, for example, can be interpreted as a functional tunnel:
Target Detection
→ Orientation
→ Pursuit
→ Capture
→ Consumption
During waking life, such tunnels support behavior.
During dreaming, similar tunnels may be replayed, recombined, reinforced, or

explored.
From this perspective, dreams may expose active functional pathways embedded within biological cognition.
Dreaming becomes less like random image generation and more like runtime activation of stored functional structures.

Why Evolution Would Preserve Dreaming

Dreaming is not free.
Brains consume substantial energy.
Sleep occupies a significant portion of an organism's life.
Any widespread and persistent biological mechanism likely provides evolutionary value.
Several possibilities have been proposed by neuroscientists and cognitive researchers:

- Memory consolidation
- Experience replay
- Skill refinement
- Threat simulation
- Emotional regulation
- Predictive preparation

The MBS perspective does not attempt to replace these explanations.
Instead, it proposes that many of them may emerge from a shared structural runtime process.
Dreaming may help:

- Strengthen triggers
- Refine retrieval pathways
- Reinforce functional tunnels
- Reorganize graph structures
- Support adaptation without external risk

If so, dreaming becomes an important mechanism for continuous structural evolution.

Threat Simulation and Survival Pressure

One particularly important possibility involves danger response.
Many animals live under constant threat.
Predators.
Competition.
Environmental hazards.
Resource scarcity.
Under such conditions, the ability to rapidly activate relevant behavioral

pathways becomes critical.

Evolution strongly favors organisms capable of:

- Detecting threats quickly
- Retrieving appropriate responses rapidly
- Executing survival behaviors efficiently

Dreaming may provide a low-risk environment for strengthening or refining such capabilities.

Whether or not this interpretation proves fully correct, it highlights a central MBS idea:

Trigger systems and retrieval systems may have been shaped by survival pressures long before advanced reasoning emerged.

From Dreaming to Cognitive Architecture

Dreaming suggests that biological cognition may not operate primarily as a symbolic reasoning engine.

Instead, it may rely heavily on:

- Trigger activation
- Retrieval mechanisms
- Graph-like organization
- Functional pathways
- Scene construction
- Runtime execution

These elements appear repeatedly across:

- Dreaming
- Memory recall
- Imagination
- Planning
- Navigation
- Skill acquisition

The recurrence of these patterns motivates the search for a unified structural framework.

This search ultimately leads to the MET Brain Structure hypothesis.

Dreaming and Cognitive Convergence

Dreaming animals also provide an intriguing clue regarding cognitive convergence.

Different individuals often develop surprisingly similar responses to similar environments.

Traditional explanations frequently emphasize learning and adaptation.

The MBS perspective adds another possibility.

If organisms share similar:

- Survival pressures
- Trigger categories
- Retrieval requirements
- Functional tunnels

then they may gradually develop similar cognitive structures.
In this view:
Similar Environment
→ Similar Triggers
→ Similar Retrieval Paths
→ Similar Functional Tunnels
→ Similar Cognition
Dreaming may participate in this process by repeatedly activating and refining these structures over time.

A Structural Interpretation

The MBS project proposes a simple but potentially powerful interpretation:
Dreaming may reveal the operation of an ancient structural runtime system.
Such a system may involve:
Trigger
→ Retrieval
→ Graph Activation
→ Tunnel Activation
→ Scene Construction
→ Learning and Adaptation
This interpretation does not claim to explain all aspects of dreaming.
Nor does it attempt to replace neuroscience.
Instead, it offers a structural lens through which diverse observations may be organized and connected.

Conclusion

Dreaming animals matter because they allow us to look beneath language, culture, mathematics, and civilization.
They reveal cognitive capabilities that appear far older than human intellectual achievements.
The existence of dreaming across species suggests that biological intelligence may depend upon deep structural mechanisms for activation, retrieval, simulation, and adaptation.
For the MET Brain Structure hypothesis, dreaming is therefore not merely a curiosity.

It is evidence.
It is a window into the possible architecture of biological cognition.
And perhaps most importantly, it reminds us that the foundations of intelligence may have been built long before intelligence learned to describe itself.